Appendix II. Update to the Low-Income Country Debt Sustainability Analysis (LIC-DSA)[1]

This Annex provides an update to the LIC-DSA prepared for the second review under the ECF arrangement (IMF Country Report n°12/332, Supp. 2, 11/16/2012). It includes updated macroeconomic information and additional information on borrowing, including on three large loans.

1. Côte d'Ivoire continues to be assessed at a moderate risk of debt distress. As in the previous assessment, the moderate rating is primarily due to Côte d'Ivoire's vulnerability to adverse shocks to exports and growth as well as to fiscal performance. This underscores the need to implement sound macroeconomic policies and ensure public investment in sound projects, while improving implementation capacity and the business climate; the above referenced LIC-DSA provides a more detailed discussion of these issues.

2. The LIC-DSA also underscores that in order to meet Côte d'Ivoire's large public investment needs without compromising debt sustainability and heightening vulnerabilities, it would be important to borrow predominantly on concessional terms and in particular to pay close attention to the impact on the debt service from new borrowing on the budget, especially over the next ten years.[2] In this regard, to contain potential future debt service, careful monitoring of PPPs would be needed to ensure that any budgetary costs are manageable. While the sustained breach of the thresholds for the PV of debt-to-GDP and debt service-to revenue under the combination shock underscores the need to monitor the implications of new debt for the budgetary burden of debt service, it is important to highlight the role of claims under the C2D which allow for some potential flexibility in the profile of debt service.[3] The share in total external debt service paid

under C2D is substantial, accounting for around 40 percent through 2015 before declining to 15 percent by 2027 (the final year of payments). However, the profile of C2D debt service can be

[1] The DSA was prepared by IMF staff in collaboration with the Côte d'Ivoire's authorities.

[2] Debt service payments in the early years of the projection period are heavily influenced by obligations on debt outstanding at end-2012, notably debt service on Cote d'Ivoire's Eurobond, repayment of a large loan from France in 2011, and repayment of ODA claims of France in exchange for development grants (see footnote 3).

[3] Claims under the C2D includes outstanding ODA debt service (principal and interest) claims of France that will be subject to a debt for development swap (C2D), amounting to$3.7 billion (15 percent of 2012 GDP). Under the C2D the existing debt service claims (as of the HIPC completion point) are being re-profiled over 15 years (2012–27), and when they are paid to France they are channeled back by France to Côte d'Ivoire through matching grants into development spending.

reviewed periodically by the authorities and the French Development Agency (AFD) to take into account Côte d'Ivoire's capacity to pay and project implementation capacity.

3. Côte d'Ivoire's external public debt amounted to $7.8 billion (31.6 percent of GDP)[4] at end-2012 (Table 1), significantly down following the attainment of the HIPC completion point from $12.6 billion (52.2 percent of GDP) at end-2011, and close to that projected in previous LIC-DSA update. The updated underlying macroeconomic framework is consistent with the ECF-supported program, and is broadly similar to the last LIC-DSA (Table 2). However, real GDP growth in 2012 was stronger than previously projected by just over 1 percent and medium-term growth rates have been increased by about 1 percent each year resulting in higher nominal GDP throughout the projection period, reflecting higher public and private investment (see main text ¶11.).

4. The baseline debt assumptions in the current LIC-DSA have changed compared to the previous LIC-DSA (Table 2), primarily reflecting (i) an upward revision in projected external borrowing (and government disbursements) with a corresponding increase in projected investment; (ii) the introduction of more up to date information on three major projects; and (iii) an increase in the program limit on new non-concessional borrowing of $200 million in 2013.[4] The higher external borrowing projections reflect the outcome of the December 2012 Consultative Group meeting which yielded higher than previously anticipated offers of financing The new information on the size of loans to finance the construction of the Soubré hydroelectric dam ($500 million), extension of the port of Abidjan (*Port Autonome Abidjan*) (about $800 million) and the extension and rehabilitation of the electricity distribution network (about $900), and the terms of the loans.[5] The impact of the new information is an increase in the size of projected new borrowing (by about $360 million) during 2014–16 and more favorable terms on the three loans as compared to the previous LIC-DSA. As in the previous LIC-DSA, during 2015–17 the main source of new borrowing would be concessional, but thereafter it gradually declines over the projection period while non-concessional borrowing steadily rises to about 3 percent of GDP near the end of the projection period.

5. The results of the external DSA confirm that Côte d'Ivoire's debt dynamics are sustainable (Figures 1 and 2; Tables 3 and 4). The stress tests reaffirm that Côte d'Ivoire continues to have vulnerabilities to macroeconomic shocks, in particular from shocks to growth, exports and

[4]The additional non-concessional borrowing is not an increase in borrowing as compared to the previous baseline but rather is a change in the composition of borrowing; it is assumed that disbursements from the additional borrowing in 2013 are made during 2013 and 2014.

[5]Some of the anticipated disbursements and associated borrowing for these projects had been included in the previous LIC-DSA. The loan for Soubré hydroelectric project was signed in January 2013 and the terms are known (and it has a grant element of 33.3 percent). For the other two projects, recent discussions between the authorities and creditors on the financing has provided updated information on the possible terms of the resulting loans; the authorities are seeking concessional terms.

fiscal performance, as illustrated most clearly by the historical average scenario and the export and combination shock stress tests. The changes in the growth projections (which feed into higher fiscal revenues) and in the baseline debt assumptions have offsetting effects, with the former implying a reduction in ratios and the latter broadly implying an increase. In total, the net impact in the baseline compared to the previous LIC-DSA is a slight improvement in the debt indicators in which the higher growth effect dominates, but a slight worsening of the debt-service indicators where the higher borrowing effect dominates. The stress tests and the alternative scenarios also yield results similar to the previous LIC-DSA.

6. With the inclusion of domestic debt in the LIC-DSA, Côte d'Ivoire's debt situation deteriorates modestly (Figure 3; Tables 5 and 6). However, public debt ratios would fall over the long run owing to the projected improvement in the macroeconomic outlook. Debt service indicators parallel those for external debt. The public debt indicators are particularly sensitive to the assumptions on GDP growth and shocks to growth. These results are similar to those in the last LIC-DSA.

Table 1. Côte d'Ivoire: Structure of External Debt

(As of end-2012, nominal)

	Million US dollars	Percent of total	Percent of GDP
Total	**7,814**	**100.0**	**31.64**
Multilateral creditors	**1,344**	**17.2**	**5.44**
IMF	791	10.1	3.20
World Bank	119	1.5	0.48
AfDB group	44	0.6	0.18
Other multilaterals	390	5.0	1.58
Official bilateral creditors	**3,851**	**49.3**	**15.59**
Paris Club	3,693	47.3	14.95
Non-Paris Club	158	2.0	0.64
Commercial creditors	**2,619**	**33.5**	10.60
Eurobond	2,604	33.3	10.54
Other commercials	15	0.2	0.06

Sources : Ivoirien authorities; and IMF staff estimates.

Table 2. Côte d'Ivoire LIC DSA Macroeconomic Assumptions: Comparison with the 2nd ECF Review LIC DSA

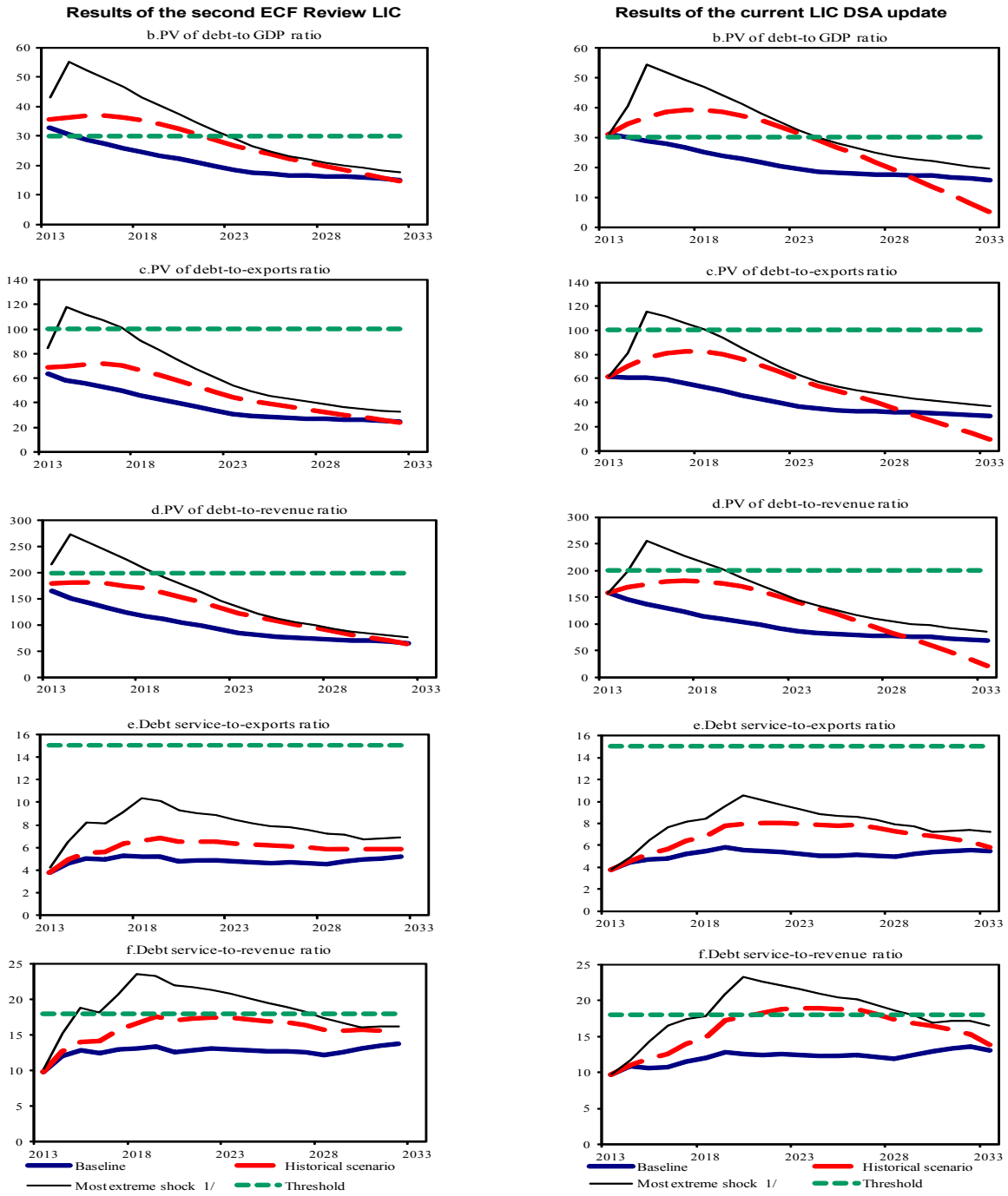
(Percent of GDP, unless otherwise indicated)

	2nd ECF Review LIC DSA						Current LIC DSA Update					
	2013	2014	2015	2018	2023	2032	2013	2014	2015	2018	2023	2032
Nominal GDP ($ billion)	26.0	28.3	30.7	39.2	57.5	130.4	28.4	31.2	34.3	44.2	65.0	160.3
Real GDP (percentage change)	8.0	7.3	7.0	6.1	5.1	5.1	8.0	8.0	8.1	7.0	5.1	5.1
Fiscal (central government)												
Revenue and grants	21.7	22.0	22.1	22.3	22.8	23.4	21.3	22.2	22.9	23.1	23.2	23.2
Primary expenditure	22.9	23.1	23.8	24.1	24.0	24.6	23.1	24.0	24.7	24.4	24.6	24.8
of which Capital expenditure	7.5	7.7	8.1	8.0	8.0	7.9	7.9	8.2	9.0	8.6	9.0	9.0
Primary fiscal balance	1.2	1.1	1.7	1.8	1.2	1.2	1.8	1.8	1.8	1.3	1.4	1.6
New external borrowing[1]	0.8	0.9	1.0	1.1	1.4	4.1	0.6	1.2	1.2	1.1	1.8	4.5
Grant element of new external borrowing (%)	10.3	30.6	24.0	2.8	-6.5	-18.9	14.5	18.6	13.8	1.9	1.3	-16.1
Balance of payments												
Exports of goods and services	51.7	51.9	51.7	52.8	59.6	61.1	50.5	49.7	47.9	47.4	53.0	55.2
Imports of goods and services	50.6	50.5	50.4	52.3	59.9	63.8	48.3	47.5	46.7	46.4	53.2	57.5
Current account, incl. official transfers	-3.4	-3.8	-3.7	-4.6	-5.3	-6.5	-2.9	-2.6	-3.6	-4.2	-5.3	-6.6
Foreign direct investment	3.1	3.1	3.2	3.3	3.6	3.7	3.0	3.0	3.0	2.9	2.5	1.7

Sources: Ivoirien authorities; and IMFstaff estimates.

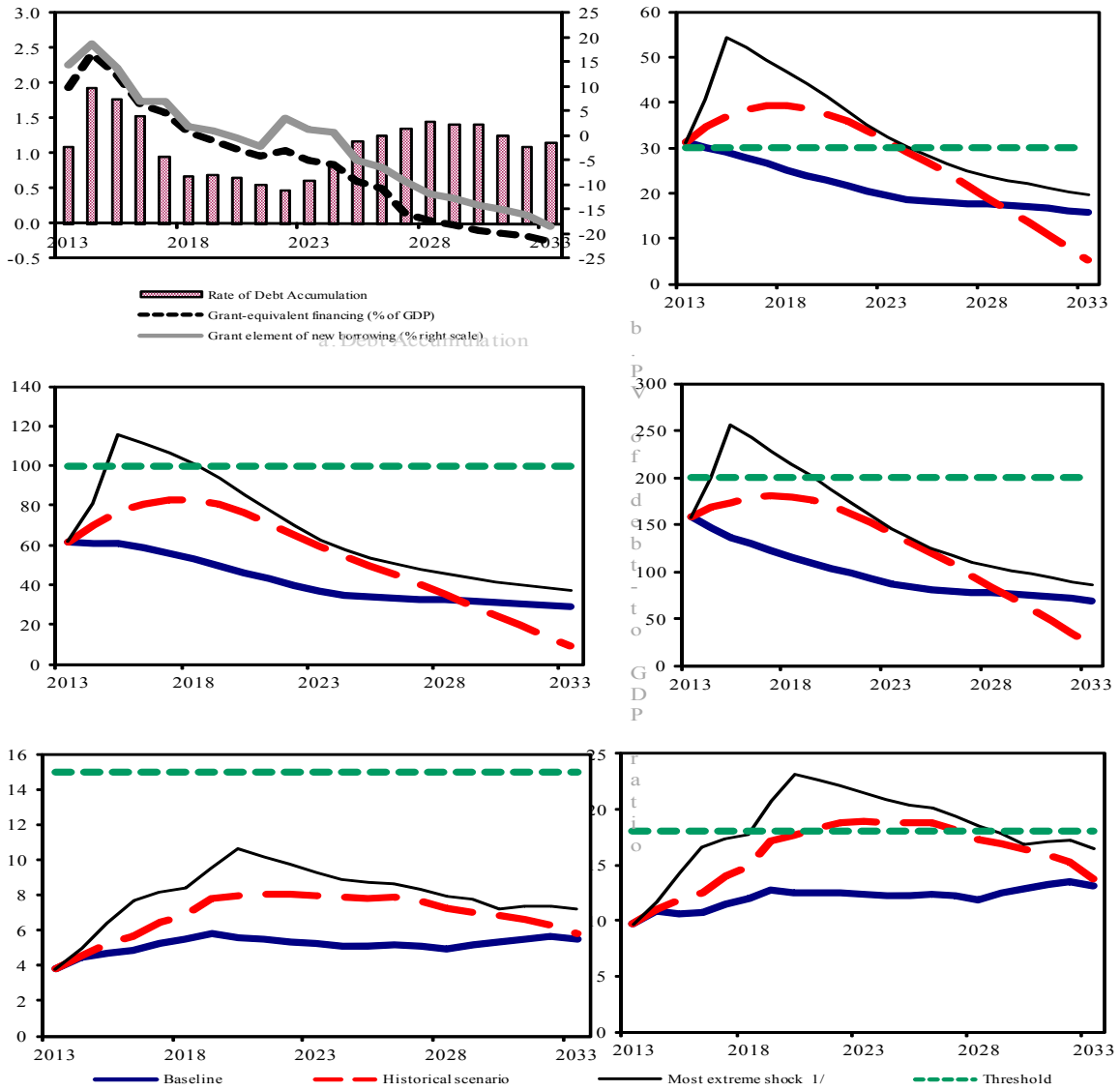
[1] Includes publicly guaranteed external borrowing.

Figure 1. Côte d'Ivoire External LIC DSA: Comparison with the Second ECF Review LIC DSA



Sources: Country authorities; and IMF staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2023 (in 2022 for the second ECF review LIC DSA). For both the LIC DSA shown, figures b, c, d, e. and f correspond to a combination shock.

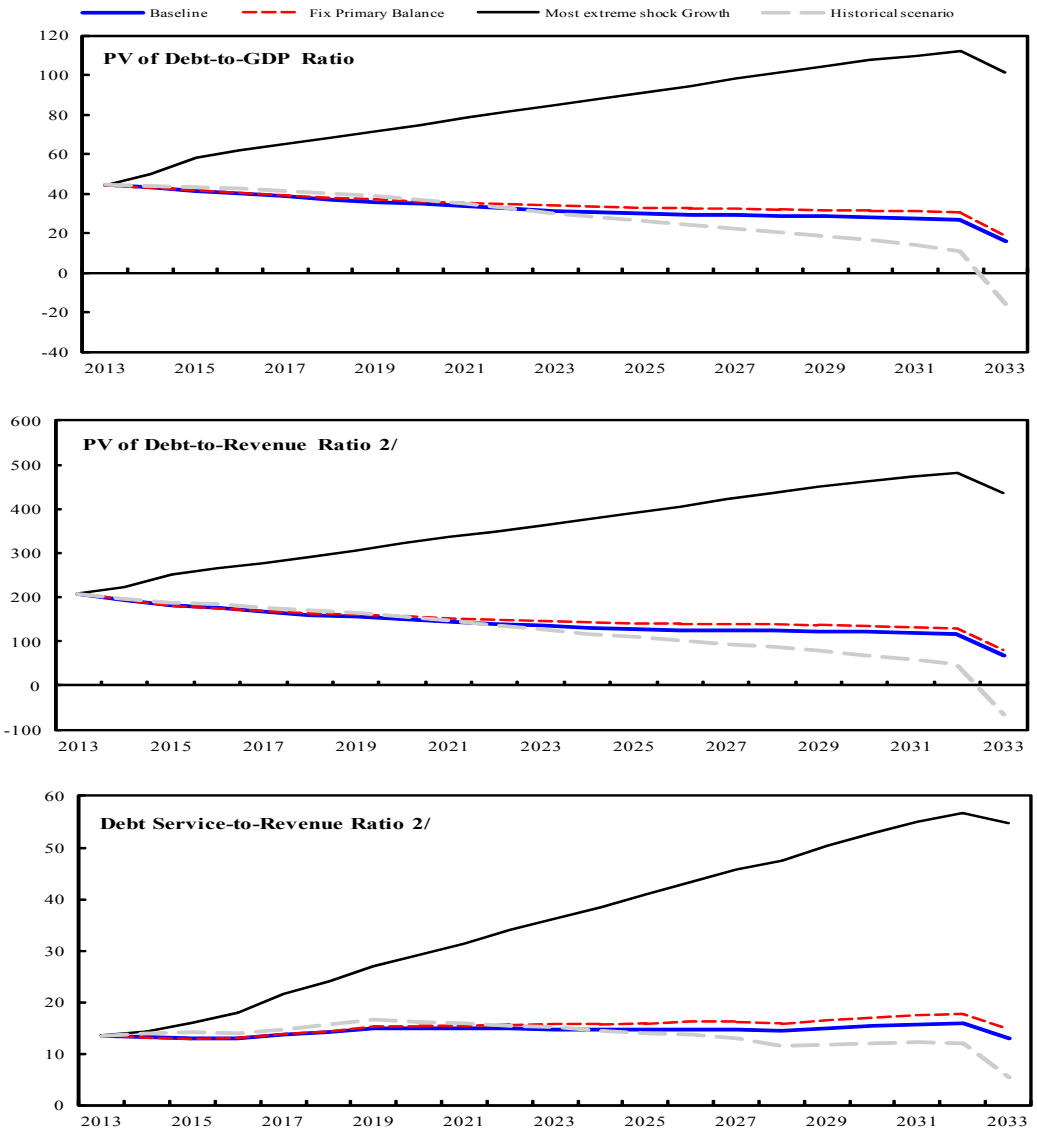
Figure 2. Côte d'Ivoire: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013–33 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2023. In figure b. it corresponds to a Combination shock; in c. to a Exports shock; in d. to a Combination shock; in e. to a Exports shock and in figure f. to a Combination shock

Figure 3. Côte d'Ivoire: Indicators of Public Debt Under Alternative Scenarios, 2013–22 [1]



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2023.
2/ Revenues are defined inclusive of grants.

Table 3. Côte d'Ivoire: External Debt Sustainability Framework, Baseline Scenario, 2010–33[1]

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average	Standard Deviation	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**76.3**	**76.9**	**51.3**			**45.6**	**43.0**	**40.3**	**37.8**	**35.1**	**32.6**		**23.3**	**15.1**	
o/w public and publicly guaranteed (PPG)	50.6	55.1	30.8			28.7	28.4	27.8	26.9	25.7	24.3		19.5	15.1	
Change in external debt	-4.9	0.6	-25.6			-5.7	-2.6	-2.7	-2.5	-2.6	-2.5		-1.5	-0.2	
Identified net debt-creating flows	-5.4	-17.5	-2.3			-3.6	-3.6	-2.5	-2.0	-1.5	-0.9		1.7	4.4	
Non-interest current account deficit	**-4.2**	**-14.9**	**-0.2**	**-5.9**	**4.0**	**1.3**	**1.3**	**2.3**	**2.5**	**2.9**	**3.1**		**4.4**	**6.0**	4.9
Deficit in balance of goods and services	-8.3	-19.8	-4.7			-2.2	-2.2	-1.2	-1.1	-1.1	-1.0		0.2	2.5	
Exports	54.2	57.7	54.8			50.5	49.7	47.9	47.5	47.3	47.4		53.0	54.3	
Imports	45.9	37.9	50.1			48.3	47.5	46.7	46.4	46.2	46.4		53.2	56.9	
Net current transfers (negative = inflow)	1.6	2.5	1.9	2.1	1.1	1.2	1.1	1.2	1.3	1.8	1.9		2.4	2.9	2.6
o/w official	-0.7	-0.1	-0.6			-1.4	-1.4	-1.4	-1.2	-1.1	-1.0		-0.6	0.0	
Other current account flows (negative = net inflow)	2.6	2.4	2.6			2.2	2.4	2.3	2.3	2.2	2.1		1.8	0.6	
Net FDI (negative = inflow)	**-1.4**	**-1.1**	**-1.8**	**-1.7**	**0.3**	**-3.0**	**-3.0**	**-3.0**	**-3.0**	**-3.0**	**-2.9**		**-2.5**	**-1.6**	-2.2
Endogenous debt dynamics 2/	**0.1**	**-1.5**	**-0.3**			**-1.9**	**-1.9**	**-1.8**	**-1.6**	**-1.4**	**-1.1**		**-0.2**	**0.1**	
Contribution from nominal interest rate	1.8	2.1	1.6			1.7	1.4	1.4	1.3	1.2	1.2		0.9	0.8	
Contribution from real GDP growth	-1.9	3.4	-7.4			-3.6	-3.3	-3.2	-2.9	-2.6	-2.3		-1.2	-0.7	
Contribution from price and exchange rate changes	0.3	-7.0	5.5			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**0.5**	**18.1**	**-23.3**			**-2.1**	**1.0**	**-0.2**	**-0.5**	**-1.2**	**-1.7**		**-3.2**	**-4.7**	
o/w exceptional financing	-2.6	-0.1	-20.3			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	54.1			48.0	44.7	41.5	38.8	36.0	33.4		23.3	15.8	
In percent of exports	...	...	98.8			95.0	90.0	86.8	81.7	76.1	70.4		43.9	29.1	
PV of PPG external debt	**...**	**...**	**33.7**			**31.1**	**30.1**	**29.0**	**28.0**	**26.5**	**25.1**		**19.4**	**15.8**	
In percent of exports	**...**	**...**	**61.5**			**61.6**	**60.5**	**60.6**	**58.8**	**56.1**	**53.0**		**36.7**	**29.1**	
In percent of government revenues	**...**	**...**	**167.1**			**157.9**	**146.7**	**136.5**	**130.2**	**122.3**	**114.9**		**87.2**	**69.0**	
Debt service-to-exports ratio (in percent)	**14.4**	**26.6**	**9.7**			**12.4**	**12.4**	**12.8**	**11.5**	**10.5**	**10.4**		**8.7**	**5.5**	
PPG debt service-to-exports ratio (in percent)	**3.4**	**5.0**	**1.9**			**3.8**	**4.5**	**4.7**	**4.8**	**5.3**	**5.5**		**5.2**	**5.5**	
PPG debt service-to-revenue ratio (in percent)	**9.5**	**19.5**	**5.2**			**9.7**	**10.8**	**10.6**	**10.7**	**11.5**	**12.0**		**12.4**	**13.1**	
Total gross financing need (Billions of U.S. dollars)	1.5	0.8	1.7			2.1	2.2	2.6	2.6	2.7	2.9		4.7	11.7	
Non-interest current account deficit that stabilizes debt ratio	0.7	-15.5	25.5			7.0	3.9	5.0	5.1	5.5	5.6		5.9	6.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	2.4	-4.7	9.8	1.8	3.7	8.0	8.0	8.1	7.8	7.5	7.0	7.7	5.1	5.1	5.2
GDP deflator in US dollar terms (change in percent)	-0.4	10.1	-6.7	6.5	9.5	6.5	1.6	1.6	1.3	1.4	1.2	2.3	3.0	5.0	3.5
Effective interest rate (percent) 5/	2.2	2.9	2.1	2.9	0.6	3.8	3.5	3.5	3.5	3.5	3.6	3.6	4.1	5.7	4.5
Growth of exports of G&S (US dollar terms, in percent)	8.6	11.6	-2.6	9.1	7.5	6.1	8.0	5.9	8.4	8.5	8.6	7.6	11.2	8.7	10.0
Growth of imports of G&S (US dollar terms, in percent)	20.0	-13.4	35.6	13.4	15.7	10.9	7.8	8.0	8.7	8.4	9.0	8.8	11.7	9.3	10.4
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	14.5	18.6	13.8	6.9	7.0	1.9	10.5	1.3	-18.5	-7.0
Government revenues (excluding grants, in percent of GDP)	19.2	14.9	20.2			19.7	20.5	21.2	21.5	21.7	21.9		22.3	22.9	22.5
Aid flows (in Billions of US dollars) 6/	0.1	0.1	0.2			0.6	0.9	0.8	0.7	0.8	0.8		1.0	0.5	
o/w Grants	0.1	0.1	0.2			0.5	0.5	0.6	0.6	0.6	0.5		0.6	0.5	
o/w Concessional loans	0.0	0.0	0.0			0.2	0.4	0.2	0.1	0.3	0.3		0.5	0.0	
Grant-equivalent financing (in percent of GDP) 7/	...	...	...			1.9	2.4	2.1	1.7	1.6	1.3		0.9	-0.3	0.4
Grant-equivalent financing (in percent of external financing) 7/	...	...	...			50.5	43.7	40.9	37.1	37.9	34.0		24.7	-7.9	11.5
Memorandum items:															
Nominal GDP (Billions of US dollars)	23.0	24.1	24.7			28.4	31.2	34.3	37.5	40.8	44.2		65.0	160.3	
Nominal dollar GDP growth	2.1	4.9	2.5			15.1	9.8	9.9	9.2	9.0	8.3	10.2	8.3	10.4	9.0
PV of PPG external debt (in Billions of US dollars)			8.5			8.8	9.4	9.9	10.4	10.8	11.1		12.6	25.0	
(PVt-PVt-1)/GDPt-1 (in percent)						1.1	1.9	1.8	1.5	0.9	0.7	1.3	0.6	1.2	1.0
Gross remittances (Billions of US dollars)	-0.5	-0.6	-0.6			-0.7	-0.8	-0.9	-1.0	-1.2	-1.3		-2.0	-4.6	
PV of PPG external debt (in percent of GDP + remittances)	...	...	34.6			31.9	30.9	29.8	28.7	27.3	25.8		20.1	16.3	
PV of PPG external debt (in percent of exports + remittances)	...	...	64.5			64.9	63.8	64.0	62.2	59.7	56.4		38.9	30.7	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.0			4.0	4.7	5.0	5.1	5.6	5.9		5.5	5.8	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Defined as grants, concessional loans, and debt relief.

7/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 4. Côte d'Ivoire: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–33

(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to GDP ratio								
Baseline	31	30	29	28	27	25	**19**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	31	35	37	38	39	39	**31**	5
A2. New public sector loans on less favorable terms in 2013-2033 2	31	32	31	31	30	29	**27**	24
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	31	32	32	31	29	28	**21**	17
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	31	36	45	43	41	39	**27**	17
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	31	33	34	33	31	29	**23**	18
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	31	34	37	36	34	32	**23**	16
B5. Combination of B1-B4 using one-half standard deviation shocks	31	41	54	52	49	47	**32**	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	31	42	41	39	37	35	**27**	22
PV of debt-to-exports ratio								
Baseline	62	61	61	59	56	53	**37**	29
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	62	70	77	81	83	83	**59**	9
A2. New public sector loans on less favorable terms in 2013-2033 2	62	64	66	65	64	62	**50**	45
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	62	60	60	58	55	52	**36**	28
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	62	81	116	111	106	101	**63**	37
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	62	60	60	58	55	52	**36**	28
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	62	69	78	75	72	68	**44**	29
B5. Combination of B1-B4 using one-half standard deviation shocks	62	81	109	105	100	95	**59**	35
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	62	60	60	58	55	52	**36**	28
PV of debt-to-revenue ratio								
Baseline	158	147	137	130	122	115	**87**	69
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	158	169	173	179	181	180	**141**	22
A2. New public sector loans on less favorable terms in 2013-2033 2	158	154	148	145	140	135	**119**	106
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	158	155	153	145	136	127	**96**	76
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	158	176	213	202	189	178	**122**	72
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	158	159	161	153	144	135	**101**	80
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	158	167	175	166	156	147	**104**	70
B5. Combination of B1-B4 using one-half standard deviation shocks	158	199	256	242	228	214	**146**	85
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	158	207	192	182	171	160	**120**	95

Table 4. Côte d'Ivoire: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013-2033 (continued)
(In percent)

Debt service-to-exports ratio								
Baseline	4	4	5	5	5	6	**5**	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	4	5	5	6	6	7	**8**	6
A2. New public sector loans on less favorable terms in 2013-2033 2	4	4	5	5	5	5	**5**	9
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	4	4	5	5	5	5	**5**	5
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	4	5	6	8	8	8	**9**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	4	4	5	5	5	5	**5**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	4	4	5	6	6	6	**6**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	4	5	6	7	8	8	**9**	7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	4	4	5	5	5	5	**5**	5
Debt service-to-revenue ratio								
Baseline	10	11	11	11	12	12	**12**	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	10	11	12	13	14	15	**19**	14
A2. New public sector loans on less favorable terms in 2013-2033 2	10	11	10	10	11	11	**12**	22
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	10	11	12	12	13	13	**14**	15
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	10	11	12	14	15	15	**18**	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	10	12	13	13	14	14	**15**	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	10	11	11	12	13	13	**15**	13
B5. Combination of B1-B4 using one-half standard deviation shocks	10	12	14	17	17	18	**22**	17
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	10	15	15	15	16	17	**17**	18
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	-20	-20	-20	-20	-20	-20	**-20**	-20

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 5. Côte d'Ivoire: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010–33

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2010	2011	2012	Average	Standard Deviation	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	66.4	71.2	45.6			42.0	41.2	40.3	39.2	37.7	36.2		31.3	15.1	
o/w foreign-currency denominated	50.6	55.1	30.8			28.7	28.4	27.8	26.9	25.7	24.3		19.5	15.1	
Change in public sector debt	0.0	4.7	-25.6			-3.6	-0.7	-0.9	-1.1	-1.5	-1.5		-0.8	-11.0	
Identified debt-creating flows	-1.0	4.3	-24.0			-1.5	-0.8	-0.7	-0.5	-0.5	-0.3		0.3	0.6	
Primary deficit	1.3	2.5	2.4	0.5	1.3	1.8	1.8	1.8	1.6	1.5	1.3	1.6	1.4	2.2	1.6
Revenue and grants	19.7	15.2	20.8			21.3	22.2	22.9	23.0	23.1	23.1		23.2	23.2	
of which: grants	0.5	0.3	0.6			1.6	1.7	1.6	1.5	1.4	1.2		0.9	0.3	
Primary (noninterest) expenditure	21.0	17.7	23.2			23.1	24.0	24.6	24.6	24.5	24.4		24.6	25.4	
Automatic debt dynamics	0.4	1.9	-6.0			-3.3	-2.5	-2.4	-2.1	-2.0	-1.6		-1.1	-1.6	
Contribution from interest rate/growth differential	-1.3	3.6	-6.5			-2.8	-2.4	-2.4	-2.2	-2.1	-1.8		-0.9	-1.2	
of which: contribution from average real interest rate	0.3	0.3	-0.1			0.6	0.7	0.7	0.7	0.7	0.7		0.7	0.0	
of which: contribution from real GDP growth	-1.6	3.3	-6.4			-3.4	-3.1	-3.1	-2.9	-2.7	-2.5		-1.6	-1.3	
Contribution from real exchange rate depreciation	1.6	-1.7	0.5			-0.5	-0.1	-0.1	0.1	0.1	0.2		...	...	
Other identified debt-creating flows	-2.6	-0.1	-20.3			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	-2.6	-0.1	-20.3			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.9	0.4	-1.6			-2.1	0.0	-0.3	-0.6	-1.0	-1.2		-1.2	-11.6	
Other Sustainability Indicators															
PV of public sector debt	...	...	48.4			44.4	42.9	41.5	40.2	38.5	37.0		31.2	15.8	
o/w foreign-currency denominated	...	...	33.7			31.1	30.1	29.0	28.0	26.5	25.1		19.4	15.8	
o/w external	...	...	33.7			31.1	30.1	29.0	28.0	26.5	25.1		19.4	15.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	4.2	6.8	4.4			4.6	4.7	4.7	4.6	4.6	4.6		4.8	5.2	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	232.8			208.2	193.4	181.6	175.2	167.2	160.2		134.8	68.1	
PV of public sector debt-to-revenue ratio (in percent)	…	…	240.3			225.3	209.5	195.5	187.4	177.9	169.3		140.1	69.0	
o/w external 3/	…	…	167.1			157.9	146.7	136.5	130.2	122.3	114.9		87.2	69.0	
Debt service-to-revenue and grants ratio (in percent) 4/	14.6	28.0	9.8			13.4	13.2	12.9	12.9	13.7	14.2		14.7	12.9	
Debt service-to-revenue ratio (in percent) 4/	15.0	28.5	10.1			14.5	14.3	13.9	13.8	14.6	15.0		15.3	13.1	
Primary deficit that stabilizes the debt-to-GDP ratio	1.3	-2.2	27.9			5.4	2.5	2.7	2.7	2.9	2.8		2.2	13.2	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	2.4	-4.7	9.8	1.8	3.7	8.0	8.0	8.1	7.8	7.5	7.0	7.7	5.1	5.1	5.2
Average nominal interest rate on forex debt (in percent)	0.8	1.9	0.9	1.7	0.6	2.8	2.4	2.5	2.6	2.8	2.9	2.7	3.9	5.6	4.3
Average real interest rate on domestic debt (in percent)	2.9	0.0	3.4	0.9	1.9	2.0	3.4	3.5	3.8	3.7	3.9	3.4	2.8	...	2.4
Real exchange rate depreciation (in percent, + indicates depreciation)	3.1	-3.2	1.0	-0.5	7.4	-1.7	...	...	...	...	...	…	...	...	…
Inflation rate (GDP deflator, in percent)	1.8	5.0	1.0	2.9	2.4	3.1	2.5	2.3	2.0	2.2	2.0	2.3	3.0	5.0	3.5
Growth of real primary spending (deflated by GDP deflator, in percent)	0.1	-0.2	0.4	0.1	0.2	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Grant element of new external borrowing (in percent)	...	...	...	…	…	14.5	18.6	13.8	6.9	7.0	1.9	10.5	1.3	-18.5	...

Sources: Country authorities; and staff estimates and projections.

1/ The public sector includes the central government and select public enterprises.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

Table 6. Côte d'Ivoire: Sensitivity Analysis for Key Indicators of Public Debt 2013–33

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	44	43	42	40	39	37	31	16
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	44	44	43	43	42	40	30	-16
A2. Primary balance is unchanged from 2013	44	43	42	40	39	38	34	19
A3. Permanently lower GDP growth 1/	44	43	43	42	42	41	44	57
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	44	50	58	62	65	68	85	101
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	44	43	41	40	38	37	31	16
B3. Combination of B1-B2 using one half standard deviation shocks	44	45	46	49	51	54	67	78
B4. One-time 30 percent real depreciation in 2014	44	56	53	52	50	48	42	31
B5. 10 percent of GDP increase in other debt-creating flows in 2014	44	55	53	51	49	47	40	22
PV of Debt-to-Revenue Ratio 2/								
Baseline	208	193	182	175	167	160	135	68
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	208	197	188	184	177	171	127	-66
A2. Primary balance is unchanged from 2013	208	193	182	176	169	164	146	81
A3. Permanently lower GDP growth 1/	208	196	187	184	180	178	190	244
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	208	222	251	266	278	291	363	435
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	208	193	181	175	167	160	135	68
B3. Combination of B1-B2 using one half standard deviation shocks	208	204	201	212	221	231	287	338
B4. One-time 30 percent real depreciation in 2014	208	250	233	225	215	207	183	132
B5. 10 percent of GDP increase in other debt-creating flows in 2014	208	246	230	223	213	204	174	96
Debt Service-to-Revenue Ratio 2/								
Baseline	13	13	13	13	14	14	15	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	13	14	14	14	15	16	15	5
A2. Primary balance is unchanged from 2013	13	13	13	13	14	14	16	15
A3. Permanently lower GDP growth 1/	13	13	13	13	14	15	19	31
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	13	14	16	18	21	24	36	55
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	13	13	13	13	14	14	15	13
B3. Combination of B1-B2 using one half standard deviation shocks	13	14	15	14	16	19	28	43
B4. One-time 30 percent real depreciation in 2014	13	15	17	18	19	21	24	29
B5. 10 percent of GDP increase in other debt-creating flows in 2014	13	13	15	19	19	17	20	17

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.